SCOPE OF WORK

Insurance Lead Management & Distribution Platform (REVRA)

Product Requirement Document & Statement of Work

Platform: MERN Stack Web Application

Version: 1.0 — Initial Release

Prepared For: Client — Insurance Operations

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Delivery Timeline: 45 Business Days

Classification: Confidential

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TABLE OF CONTENTS

1	Executive Summary	3
2	Project Overview & Business Context	3
3	Product Vision & Strategic Objectives	4
4	Success Metrics & KPIs	4
5	System Architecture	5
6	User Roles & Access Control	5
7	Module 1 — Lead Management System	6
8	Module 2 — Lead Distribution Engine	7
9	Module 3 — Subscription Management	8
10	Module 4 — Agent Dashboard & Performance	9
11	Module 5 — Follow-Up & Reminder System	10
12	Module 6 — Admin Control Panel	10
13	Module 7 — Analytics & Reporting	11
14	Meta Ads API Integration	12
15	Security Architecture	12
16	Technology Stack	13
17	Database Design	13
18	API Design Overview	14
19	Platform Scope & Responsive Design	14
20	Project Timeline — 45-Day Plan	15
21	Deliverables	15
22	Out of Scope	16
23	Risks, Assumptions & Dependencies	16

01 EXECUTIVE SUMMARY

This document constitutes the formal Scope of Work (SOW) and Product Requirement Document (PRD) for the design, development, and deployment of an Insurance Lead Management and Distribution Platform. The system is purpose-built for insurance agencies and platform operators who require an efficient, automated, and scalable solution to capture, process, and distribute insurance leads sourced from Meta Ads campaigns.

The platform will be developed using the MERN stack (MongoDB, Express.js, React.js, Node.js) and will consist of two primary interfaces: an Admin Panel for platform operators and a web-based Agent Application for insurance agents. The system is designed to handle the complete lead lifecycle — from initial capture through Meta Ads API integration, through intelligent automated distribution, to final conversion tracking — while providing powerful analytics and a subscription-based monetization layer.

This document covers all functional and non-functional requirements, system architecture, module specifications, API design, database schema, security protocols, deployment strategy, and a phased 45-day project timeline. It is the single authoritative reference for all development activities and client-agency communications throughout the project lifecycle.

02 PROJECT OVERVIEW & BUSINESS CONTEXT

The insurance industry relies heavily on lead generation and timely follow-up to drive policy sales. Many insurance operators currently manage leads through manual, fragmented processes — spreadsheets, WhatsApp groups, and informal assignments — leading to delays, missed leads, duplicate contacts, and an inability to track performance or conversion outcomes. This platform is engineered to eliminate all of those inefficiencies through a centralized, automated, data-driven system.

2.1 Core Problem Statements

- No centralized system to capture and manage leads from Meta Ads campaigns in real time.
- Manual lead distribution leads to unequal workloads, missed assignments, and poor accountability.
- Agents have no visibility into their own performance metrics, targets, or follow-up schedules.
- No subscription or quota management system to monetize and control lead access for agents.
- Admin has limited ability to monitor agent activity, platform-wide conversion rates, or revenue.
- Absence of a structured CRM-like pipeline means high-value leads are lost after the first contact attempt.
- No audit trail of lead activity makes disputes between agents and management impossible to resolve.

2.2 Proposed Solution

The proposed platform solves all of the above by delivering an end-to-end Lead Management and Distribution System with the following core capabilities:

- Automated, real-time lead ingestion from Meta Ads via API integration (webhook and polling modes).
- Intelligent, rule-based lead distribution engine with configurable allocation strategies.
- Full CRM-like lead pipeline with status tracking, notes, follow-ups, and immutable activity logs.
- Subscription management system with tiered plans, lead quotas, and automated enforcement.
- Personalized real-time dashboards for both the admin and individual agents.
- Target setting and performance tracking per agent with visual progress indicators.

- Follow-up scheduling with overdue flagging and dashboard reminders.
- Comprehensive analytics with export capabilities for both operational and revenue reporting.

Client Type	Insurance Agency / Platform Operator
Target Users	Platform Admin (1), Insurance Agents (Multiple concurrent users)
Core Tech Stack	MERN — MongoDB, Express.js, React.js, Node.js
Primary Lead Source	Meta Ads API (Facebook & Instagram Lead Ad Forms)
Delivery Model	SaaS-style web platform with subscription-based agent access
Project Timeline	45 Business Days from confirmed project kick-off date

03 PRODUCT VISION & STRATEGIC OBJECTIVES

The platform vision is to build a high-performance lead distribution ecosystem that maximizes lead utilization, agent productivity, and conversion efficiency while creating a sustainable, subscription-driven revenue model for the platform operator. The six strategic objectives below define what success looks like beyond technical delivery.

Maximize Lead Utilization: Every lead captured from Meta Ads must be assigned to an active, eligible agent with zero waste and minimal processing delay, targeting assignment within 5 seconds of receipt into the system.

Empower Agents with Actionable Data: Each agent receives a personalized dashboard showing their leads, conversion rates, follow-up obligations, and progress against personal targets — enabling data-driven, disciplined selling behavior.

Create a Monetizable, Scalable Platform: Enable the admin to offer tiered subscription plans to agents, controlling lead access volumes and generating recurring, predictable subscription revenue from the agent network.

Automate All Repetitive Operations: Eliminate manual work from lead assignment, status routing, follow-up reminders, and performance reporting — freeing admin and agents to focus exclusively on high-value activities.

Establish Full Accountability: Create a transparent, tamper-evident record of every lead — who it was assigned to, when it was contacted, all notes added, and the final outcome — eliminating disputes and information gaps.

04 SUCCESS METRICS & KEY PERFORMANCE INDICATORS

The following KPIs define what a successful platform delivery looks like and will be used to evaluate platform performance during UAT and the 30-day post-launch warranty period.

KPI	Target	Measurement Method
	Less than 5 seconds from receipt	System log timestamps
Agent Engagement Rate	Greater than 80% of leads actioned	Status change tracking
	100% within configured quota rules	Distribution audit log
Conversion Tracking Coverage	100% of assigned leads tracked to outcome	Lead pipeline records
	Real-time cutoff on plan expiry	Automated rule engine
System Uptime	99.9% availability	Uptime monitoring tools
	Under 2 seconds	Performance testing suite
	Less than 1% post-filtering	Deduplication log reports

05 SYSTEM ARCHITECTURE

The platform is designed as a Modular Monolithic Architecture — providing the operational simplicity of a single deployable unit while maintaining clean internal module boundaries that allow future decomposition into microservices as scale demands. The architecture is organized into five primary layers:

Frontend Layer: React.js single-page application (SPA) styled with Tailwind CSS. Separate route-protected views for Admin Panel (desktop-optimized) and Agent Application (mobile-responsive). State managed via React Context API or Redux Toolkit.

Backend Layer: Node.js runtime with Express.js HTTP framework providing a structured RESTful API. Organized into service modules: Authentication, Leads, Distribution Engine, Subscriptions, Users, and Analytics. Business logic is fully decoupled from route handlers.

Database Layer: MongoDB document database chosen for its natural fit with lead data structures that may have variable fields. Mongoose ODM provides schema validation and type enforcement. Critical fields are indexed for high-read query performance.

Integration Layer: Meta Lead Ads API connector supporting both webhook-based (push) and polling-based (pull) lead ingestion modes. Duplicate detection and validation middleware applied before any lead enters the distribution pipeline.

Infrastructure Layer: Deployed to cloud VPS (AWS EC2 or equivalent). Nginx reverse proxy handles HTTPS termination and static asset serving. PM2 manages the Node.js process with auto-restart and optional clustering. CI/CD pipeline enables zero-downtime deployments.

End-to-End Data Flow: Meta Ads generates leads > Backend fetches via API > Validation & deduplication > Distribution Engine assigns to eligible agents > Agents receive, update status, add notes > Activity logs written > Admin and Agent dashboards query analytics in real time.

06 USER ROLES & ACCESS CONTROL

The platform implements a two-tier Role-Based Access Control (RBAC) model. All access is governed by JWT authentication with role claims embedded in the token payload. Tokens have a 24-hour expiry with a secure refresh token mechanism.

Role	Description	Key Permissions
Admin	Platform owner/operator with full system access.	Create/manage agents, manage subscriptions and plans, view all leads in the system, configure distribution strategy, access full analytics, override lead assignments, manage reserve lead pool.
Agent	Insurance sales agent operating within their subscription quota.	View own assigned leads, update lead status and notes, schedule follow-ups, set personal performance targets, view own dashboard and conversion metrics.

All API endpoints are protected by JWT middleware. Role-specific guards prevent agents from accessing admin-only routes. Agents cannot access other agents' lead records under any circumstances.

07 MODULE 1 — LEAD MANAGEMENT SYSTEM

The Lead Management System is the operational core of the platform. It provides both admin and agents with complete visibility and control over individual leads across their full lifecycle — from initial assignment through to final disposition.

7.1 Lead Data Schema

Each lead record contains the following standardized fields capturing Meta Ads data alongside platform-specific tracking information:

Field	Type	Description
Name	String	Full name of the lead as submitted on the Meta Ad form.
Phone Number	String	Primary contact number, stored with country code.
Email Address	String	Email as provided. Optional depending on the ad form configuration.
Location	String	City or region captured from ad form submission.
Lead Source	String	Set to 'Meta Ads' with sub-source tracking including campaign ID and ad set ID.
Assigned To	ObjectId (ref: User)	Agent the lead is assigned to. Null for unassigned or reserve leads.
Status	Enum	Current stage in the pipeline (New / Contacted / Hot / Warm / Follow-Up / Not Interested / Converted).
Notes	Array of Objects	Each note: { author (ObjectId), content (String), timestamp (Date) }.
Follow-Ups	Array of Objects	Each follow-up: { scheduledDate, note, isCompleted, completedAt }.
Activity Log	Array of Objects	Immutable record of every status change and update with actor and timestamp.
Meta Lead ID	String (unique)	Original ID from Meta Ads API used for deduplication.
Created At / Updated At	Timestamp	UTC timestamps of record creation and last modification.

7.2 Lead Status Pipeline

Leads progress through a defined set of statuses representing their position in the sales lifecycle. Every status change is recorded in the immutable activity log with the acting user's identity and timestamp.

Status	Meaning	Who Can Set
New	Freshly assigned. No contact made.	System (automatic on assignment)
Contacted	Agent has made at least one contact attempt.	Agent / Admin
Hot	Lead is highly interested and likely to convert soon.	Agent / Admin
Warm	Lead shows interest but needs further nurturing.	Agent / Admin

Follow-Up	Specific callback or follow-up scheduled for a future date.	Agent / Admin
Not Interested	Lead has confirmed disinterest. Pipeline closed.	Agent / Admin
Converted	Lead has successfully purchased a policy.	Agent / Admin

7.3 Lead Pipeline View (CRM Interface)

The agent lead interface provides either a Kanban-style column view or a sortable list view with the following capabilities:

- Filter by any combination of pipeline status values.
- Filter by date range: created date, last updated date, or follow-up scheduled date.
- Filter by assigned agent (admin view only, to examine individual agent pipelines).
- Full-text search across lead name, phone number, and email address.
- Sort by newest, oldest, last activity, or current status.
- Pagination with configurable page sizes (20, 50, 100 records per page).
- Quick-action buttons on each lead card: update status, add note, schedule follow-up — without opening the full lead detail view.
- Export the current filtered view to CSV or Excel for offline reporting.

08 MODULE 2 — LEAD DISTRIBUTION ENGINE

The Lead Distribution Engine is the most technically critical component of the platform. It is responsible for automatically and fairly assigning every incoming lead to eligible agents based on a configurable rule set. The engine operates in near real-time, targeting a lead assignment within 5 seconds of receipt.

8.1 Eligibility Criteria

Before assigning a lead to an agent, the distribution engine validates all three of the following conditions:

- The agent's subscription status must be 'Active'. Expired or cancelled subscriptions immediately disqualify the agent from receiving new leads.
- The agent's remaining lead quota (plan limit minus leads already received in the current subscription period) must be greater than zero.
- The agent's account must be active and not suspended or deactivated by the admin.

8.2 Distribution Strategies

The admin can configure one of the following strategies from the Admin Panel. Strategy changes take effect immediately for all subsequent lead assignments.

Round Robin: Leads are assigned to eligible agents in a strict circular rotation. Agent A receives lead 1, Agent B receives lead 2, Agent C receives lead 3, then the cycle repeats. This guarantees perfectly equal distribution regardless of individual conversion rates or seniority.

Weighted Distribution: Agents are assigned a priority tier (Premium, Standard, Basic) by the admin. Higher-tier agents receive a proportionally larger share of leads. For example, a Premium agent may receive 3 leads for every 1 lead assigned to a Basic-tier agent.

Availability-Based: The engine tracks agent engagement signals (last login time, recency of lead updates) and biases assignment toward agents who have been most recently active. This ensures leads flow toward engaged agents rather than dormant accounts.

8.3 Reserve Leads System

The admin can configure a reserve percentage of all incoming leads to be withheld from automatic distribution and held in a managed Reserve Lead Pool. This gives the admin strategic control to allocate high-value leads to specific agents or new subscribers.

Configuration	Example	System Behavior
Distribution Percentage	80%	80 out of every 100 incoming leads are auto-distributed to eligible agents via the configured strategy.
Reserve Percentage	20%	20 out of every 100 leads are withheld and placed in the Admin Reserve Pool — unassigned.
Reserve Assignment	Manual	Admin reviews reserve leads in the Admin Panel and assigns them individually to specific agents.

Reserve Release	On demand	Admin can release the entire reserve pool to the distribution engine at any time with a single action.

8.4 Duplicate Lead Detection

Before any lead is stored or distributed, the system performs a three-layer duplicate check: Meta Lead ID uniqueness, phone number matching, and email address matching against all existing records. If a duplicate is detected, the system either silently rejects it or flags it for admin review based on the configured behavior. Agents are never assigned leads they have previously received.

09 MODULE 3 — SUBSCRIPTION MANAGEMENT SYSTEM

The Subscription Management System is the monetization engine of the platform. It allows the admin to define tiered access plans, assign them to agents, and enforce quota and access controls automatically based on plan parameters.

9.1 Subscription Plan Structure

Plan Type	Duration	Lead Quota	Access Level	Managed By
Basic Monthly	30 days	Configurable per plan	Standard lead access, dashboard, notes, follow-ups.	Admin
Professional 6-Month	180 days	Configurable per plan	All Basic features plus priority in weighted distribution.	Admin
Premium Annual	365 days	Configurable per plan	All Pro features plus weighted distribution priority and reserve lead access.	Admin

All plan parameters (name, duration, quota, pricing notes) are fully configurable by the admin. New plan types can be created at any time. Existing plan parameters cannot be changed retroactively for active subscriptions.

9.2 Subscription Lifecycle & Enforcement

- Admin creates subscription plan templates in the Admin Panel with defined duration and lead limits.
- Admin assigns a plan to a specific agent — the subscription start date is set at assignment.
- The system calculates the expiry date automatically and begins tracking the consumed lead quota.
- As leads are assigned to the agent, the consumed quota counter increments in real time.
- When the agent's quota reaches the plan limit, the distribution engine stops assigning new leads until the plan is renewed.
- When the expiry date passes, the subscription status automatically changes to 'Expired' and all lead assignment halts immediately.
- Expired agents retain complete read-only access to their historical lead data and activity logs.
- Admins receive a dashboard notification for agents approaching expiry (configurable threshold, default: 7 days).
- Admins can manually renew, upgrade, downgrade, or cancel any agent subscription at any time.

10 MODULE 4 — AGENT DASHBOARD & PERFORMANCE TRACKING

The Agent Dashboard is the primary interface for insurance agents — designed to provide instant, actionable visibility into their workload, performance metrics, and daily priorities. Every element of the dashboard is personalized to the individual agent; no agent can view another agent's data.

10.1 Dashboard Metrics & Summary Cards

Metric	Description
Total Leads Received	Cumulative count of all leads ever assigned to the agent, across all time.
Leads This Month	Leads received in the current calendar month with trend indicator vs. prior month.
Active Leads	Count of leads currently in New, Contacted, Hot, Warm, or Follow-Up status.
Converted Leads	Total leads marked as Converted, shown for all-time and the current subscription period.
Pending Follow-Ups	Count of follow-ups scheduled for today or with a past due date (overdue).
Conversion Rate	Calculated as: Converted Leads / Total Leads Received x 100%. Updated in real time.
Quota Remaining	Leads remaining in the current subscription before the plan limit is reached.
Subscription Expiry	Days remaining on the current active plan with a visual warning below 7 days.

10.2 Target Setting & Progress Tracking

Agents can define personal performance targets on a daily, weekly, or monthly basis. The system tracks actual performance against these targets and displays real-time progress.

- Agent sets a numeric conversion target for a selected time period (daily / weekly / monthly).
- Dashboard displays a progress bar: Converted Leads / Target x 100%.
- Color-coded status indicator: Green (on track), Amber (at risk — below 70% of target pace), Red (behind — below 50%).
- Full target history is retained so agents can review performance trends over multiple periods.
- Admin can view all agents' target progress from the Admin Analytics Panel.
- Targets are voluntary and self-managed by agents; admins can view but cannot override agent targets.

10.3 Activity Feed

A chronological activity feed on the agent dashboard displays recent events: newly assigned leads, status updates completed, follow-ups due today, overdue follow-ups, and recent conversions. This eliminates the need to search through the full lead list to understand what requires immediate attention.

11 MODULE 5 — FOLLOW-UP & REMINDER SYSTEM

The Follow-Up System ensures that no lead is forgotten after the initial contact attempt. Agents can schedule structured follow-up actions on any lead, and the system actively tracks overdue and upcoming follow-ups with clear visual indicators throughout the platform.

- Any agent can add a follow-up entry to any lead they manage, specifying a scheduled date and time.
- Each follow-up entry includes a free-text field for the agent to note the intended next action (e.g., 'Call back to discuss premium breakdown and policy options').
- The lead status is automatically set to 'Follow-Up' when a future follow-up is scheduled and no higher-priority status is active.
- The Agent Dashboard displays a dedicated 'Follow-Ups Today' section listing all leads with follow-ups due on the current date.
- Overdue follow-ups — those with a past due date that have not been marked as completed — are flagged prominently in red throughout the platform.
- Complete follow-up history is preserved in the lead's immutable activity log, recording all past scheduled dates, notes, and completion status.
- Browser-based notification alerts for upcoming and overdue follow-ups (Phase 2 enhancement — post-launch roadmap).
- Admin has visibility into overdue follow-ups across all agents from the Admin Panel.

12 MODULE 6 — ADMIN CONTROL PANEL

The Admin Control Panel is a dedicated desktop-optimized web interface providing the platform operator with comprehensive oversight and control of all system functions. It is exclusively accessible by users with the Admin role.

12.1 User Management

- Create new agent accounts with full profile: name, email, phone, and role assignment.
- Edit existing agent profiles and contact details at any time.
- Suspend or deactivate agent accounts — immediately halts lead assignment upon activation.
- View each agent's complete profile summary: subscription status, total leads received, conversion rate.
- Manually override the distribution engine to assign a specific lead directly to any agent.
- Reset agent passwords and manage authentication credentials.

12.2 Lead Management (Admin View)

- View the complete lead database with full filtering, sorting, and search capabilities.
- View assignment status, history, and all activity for any lead in the system.
- Reassign any lead from one agent to another at any time.
- Delete or archive stale, invalid, or test leads.
- Access and manage the Reserve Lead Pool — leads withheld from auto-distribution.
- Export the full or filtered lead database to CSV or Excel.

12.3 Subscription Administration

- Create, edit, and archive subscription plan templates with custom durations and quota limits.
- Assign any plan to any agent with a specified start date.
- View all active subscriptions, quota consumption levels, and expiry dates on a single screen.
- Manually extend, modify, or cancel active subscriptions.
- Receive expiry alert notifications for agents approaching the end of their subscription period.

12.4 Distribution Configuration

- Select and activate the distribution strategy: Round Robin, Weighted, or Availability-Based.
- Set the global reserve lead percentage (0–100% of incoming leads).
- Configure premium agent tiers and their respective lead weighting for weighted distribution.
- View the full distribution audit log showing how, when, and why each lead was assigned.
- Release reserve leads to the auto-distribution engine in bulk or individually.

13 MODULE 7 — ANALYTICS & REPORTING

The Analytics module provides real-time and historical performance data for both admin and agents. Data is visualized through charts, summary cards, and tabular reports built with Recharts or Chart.js within the React frontend.

13.1 Admin Analytics Dashboard

- Total leads received — all time, this month, and this week — with trend comparison chart.
- Leads distributed versus leads held in reserve, shown as a proportional bar or donut chart.
- Platform-wide conversion rate: Converted Leads / Total Distributed Leads x 100%.
- Agent performance comparison table: leads assigned, converted, pending, and conversion rate per agent.
- Subscription revenue summary: active plan count, plan type breakdown, and upcoming renewals calendar.
- Lead source analysis: which Meta Ads campaigns and ad sets are generating the highest lead volumes.
- Lead status distribution chart showing the current pipeline health across the entire platform.
- Top-performing agents by conversion rate — ranked leaderboard.
- Overdue follow-ups report: full list of all unactioned overdue follow-ups across all agents.

13.2 Agent Analytics View

- Personal lead funnel: how many of my leads are in each pipeline status.
- Daily, weekly, and monthly leads received trend chart with period-over-period comparison.
- Personal conversion rate over time — enabling agents to track improvement or identify declines.
- Target achievement visualization: actual versus target conversions for each configured period.
- Follow-up compliance rate: percentage of scheduled follow-ups actioned by or before the due date.

13.3 Export & Reporting

- All data tables across the platform support CSV and Excel export with current filter state applied.
- Admin can generate a comprehensive platform report covering any custom date range.
- Reports are generated server-side and served for download within seconds.

14 META ADS API INTEGRATION

Meta Ads (Facebook and Instagram) is the sole lead source for this platform at launch. The integration connects the platform backend to the Meta Lead Ads API, enabling automatic ingestion of leads submitted through Meta ad lead forms directly into the platform database.

14.1 Integration Approach

- The client must provide a valid Meta Business Account, App credentials (App ID and App Secret), and a Page Access Token before backend development begins.
- Two ingestion modes will be implemented: Webhook-based (Meta pushes a notification immediately when a lead is submitted — preferred mode) and Polling-based (backend fetches new leads on a scheduled interval — fallback mode).
- All raw lead data received from Meta is persisted to the database in its original form before any processing, creating a complete audit trail.
- The integration will capture all standard Lead Ad form fields: name, phone, email, and any custom questions configured in the ad form.

14.2 Lead Ingestion Pipeline

- Step 1 — Receipt: Meta sends lead data to the platform webhook endpoint, or the poller fetches it on schedule.
- Step 2 — Validation: Required fields (name, phone) are checked. Incomplete leads are flagged for admin review rather than silently dropped.
- Step 3 — Deduplication: System checks Meta Lead ID, phone number, and email against all existing records. Confirmed duplicates are rejected.
- Step 4 — Storage: Valid, unique leads are stored in the Leads collection with status set to 'Unassigned'.
- Step 5 — Distribution: The distribution engine triggers immediately to assign the lead to an eligible agent or hold it in reserve.

Important: Lead quality and volume are entirely dependent on the client's Meta Ads campaigns, targeting, and ad creative. The platform processes all leads received regardless of quality. AI-based lead quality scoring is a future roadmap item and is not included in this scope.

15 SECURITY ARCHITECTURE

Authentication: JWT-based stateless authentication. Access tokens carry user ID and role claim, expire in 24 hours. Paired with secure, HTTP-only cookie refresh tokens for session renewal.

Authorization: RBAC enforced at the Express middleware layer. Admin-only routes are inaccessible to agent tokens. Agents are restricted to their own lead data only — cross-agent data access is architecturally impossible.

API Security: All request bodies are validated and sanitized using express-validator. Parameterized Mongoose queries prevent NoSQL injection. Rate limiting applied to authentication endpoints to prevent brute-force attacks.

Data Security: Passwords stored exclusively as bcrypt hashes with a minimum work factor of 12. Database connection strings and secret keys stored in environment variables — never in source code. MongoDB access restricted to application server IPs via network firewall.

Transport Security: HTTPS enforced across all routes via SSL/TLS certificate managed through Let's Encrypt with automated renewal. Nginx configured to redirect all HTTP traffic to HTTPS with HSTS headers.

Infrastructure Security: VPS access via SSH key pairs only — password-based SSH authentication is disabled. Server firewall allows only ports 80, 443, and 22. Automated daily database backups retained for 14 days with off-server storage.

16 TECHNOLOGY STACK

Layer	Technology	Purpose / Notes
Frontend Framework	React.js	Component-based SPA for Admin Panel and Agent Application.
Styling	Tailwind CSS	Utility-first CSS for rapid responsive UI development.
Backend Runtime	Node.js	JavaScript server runtime for the application layer.
HTTP Framework	Express.js	Lightweight, unopinionated REST API framework.
Database	MongoDB	NoSQL document store for lead, user, and subscription data.
ODM	Mongoose	Schema definition, validation, and query building for MongoDB.
Authentication	JSON Web Tokens (JWT)	Stateless auth with embedded role claims.
Password Security	bcrypt	Industry-standard adaptive hashing algorithm.
Lead Ingestion	Meta Lead Ads API	Webhook and polling-based lead capture from Meta campaigns.
State Management	React Context API / Redux	Frontend application state management.
Charting	Recharts or Chart.js	Analytics and dashboard data visualizations.
Web Server	Nginx	Reverse proxy, HTTPS termination, and static file serving.
Process Manager	PM2	Node.js process management with auto-restart and log handling.
Hosting	AWS EC2 / Cloud VPS	Production server infrastructure (client-provisioned).
Version Control	Git + GitHub / GitLab	Source code versioning and release management.

17 DATABASE DESIGN

The MongoDB database is organized into the following primary collections. Inter-collection relationships are managed via ObjectId references. Indexes are applied on all high-frequency query fields to ensure performance as data volumes grow.

Users Collection

Fields: _id, name, email (unique), phone, passwordHash, role (admin | agent), isActive (Boolean), createdAt, updatedAt.

Indexes: email (unique index), role, isActive.

Leads Collection

Fields: _id, name, phone, email, location, source, metaLeadId (unique), status (enum), assignedTo (ObjectId ref: Users), notes (Array), followUps (Array), activityLog (Array), createdAt, updatedAt.

Indexes: metaLeadId (unique), assignedTo, status, createdAt.

Subscriptions Collection

Fields: _id, userId (ObjectId ref: Users), planId (ObjectId ref: Plans), status (active | expired | cancelled), leadsConsumed (Integer counter), leadsLimit (Integer), startDate, endDate, createdAt.

Indexes: userId, status, endDate.

Plans Collection

Fields: _id, name, durationDays (Integer), leadLimit (Integer), description, isActive (Boolean), createdAt.

Indexes: isActive.

ActivityLogs Collection

Fields: _id, entityType (lead | user | subscription), entityId (ObjectId), action (String), performedBy (ObjectId ref: Users), details (Object), timestamp.

Indexes: entityId, performedBy, timestamp.

18 API DESIGN OVERVIEW

The backend exposes a RESTful API organized into logical endpoint groups. All endpoints return JSON responses with standard HTTP status codes. Authentication is enforced via Bearer JWT in the Authorization request header. Role guards protect admin-only endpoints.

Authentication Endpoints

POST /auth/login
POST /auth/register
POST /auth/refresh-token
POST /auth/logout

Leads Endpoints

GET /leads (list — filterable, paginated)
GET /leads/:id (full lead detail)
POST /leads (manual create — admin)
PUT /leads/:id/status (update status)
POST /leads/:id/notes (add note)
POST /leads/:id/follow-up (schedule follow-up)
POST /leads/:id/assign (manual assign — admin only)

Users Endpoints

GET /users (list — admin only)
GET /users/:id
POST /users (create agent — admin)
PUT /users/:id (edit profile)
PATCH /users/:id/status (activate / suspend)

Subscriptions Endpoints

GET /subscriptions (all active — admin view)
GET /subscriptions/my (agent's own subscription)
POST /subscriptions (assign plan — admin)
PUT /subscriptions/:id (modify — admin)

Plans Endpoints

GET /plans
POST /plans (create — admin)
PUT /plans/:id (edit — admin)
DELETE /plans/:id (archive — admin)

Analytics Endpoints

GET /analytics/admin (admin dashboard data)
GET /analytics/agent (agent dashboard data)
GET /analytics/export (CSV / Excel download)

Distribution Endpoints

POST /distribution/configure (set strategy — admin)
GET /distribution/log (assignment history — admin)
POST /distribution/release-reserve (release reserves — admin)

19 PLATFORM SCOPE & RESPONSIVE DESIGN

Interface	Primary Device Target	Design Approach	Deployment
Admin Panel	Desktop (1280px+)	Desktop-first layout optimized for data-dense tables, multi-column forms, charts, and management controls requiring large screen real estate.	React SPA served via Nginx. Role-protected routes.
Agent Application	Mobile + Desktop	Mobile-responsive using Tailwind breakpoints. Agent workflow designed for use on smartphone in the field. Key actions accessible within 2 taps from the dashboard.	Same React SPA build. Routes protected by agent role JWT claims.

A dedicated native mobile application (Android or iOS) is explicitly out of scope for this engagement. The mobile-responsive web application provides sufficient mobile functionality for agents in the field and eliminates app store distribution complexity. Native mobile app development may be scoped as a separate future engagement upon request.

20 PROJECT TIMELINE — 45-DAY DELIVERY PLAN

Phase	Days	Duration	Key Deliverables & Milestones
Phase 1 — Discovery & Planning	Day 1–5	5 days	Requirement finalization, technical architecture sign-off, database schema design, API contract definition, UI wireframe approval by client.
Phase 2 — UI/UX Design	Day 6–10	5 days	High-fidelity mockups for Admin Panel and Agent App. Design system definition (colors, typography, component library). Client sign-off required before development begins.
Phase 3 — Backend Development	Day 11–25	15 days	Database setup, JWT auth system, all CRUD APIs, Distribution Engine, Meta Ads integration (webhook + polling), Subscription enforcement logic, Analytics query layer.
Phase 4 — Frontend Development	Day 11–30	20 days (parallel with backend)	Admin Panel UI, Agent App UI, API integration, Lead Pipeline view, Dashboard components, Subscription Management UI, Analytics charts.
Phase 5 — Integration & Testing	Day 31–40	10 days	End-to-end integration testing, distribution engine stress testing, Meta API integration testing, User Acceptance Testing (UAT) with client, bug fixing, performance optimization.
Phase 6 — Deployment & Handover	Day 41–45	5 days	Production server setup, CI/CD pipeline, domain & SSL configuration, final QA on live environment, technical documentation delivery, 60-90 min client walkthrough session.

Timeline Assumptions: The 45-day schedule requires client feedback within 48 hours of milestone deliveries, Meta Ads API credentials provided before Day 11, and no scope changes after Phase 1 sign-off. Any post-sign-off scope modifications are subject to a formal change order process with revised timeline and cost assessment.

21 DELIVERABLES

The following deliverables are committed as part of this scope of work and are included in the agreed project price unless otherwise noted.

#	Deliverable	Description
1	**Admin Panel**	Fully functional desktop-optimized web application with user management, subscription management, lead oversight, distribution configuration, and analytics.
2	**Agent Web Application**	Mobile-responsive web application with lead pipeline, lead detail view, follow-up management, target tracking, and personal performance dashboard.
3	**Backend API**	Complete RESTful API with all endpoints implemented, secured with JWT and RBAC, and tested. Full business logic for distribution, subscription enforcement, and analytics.
4	**Meta Ads Integration**	Configured and tested Meta Lead Ads API integration with webhook-based ingestion, deduplication pipeline, and validation logic.
5	**Database Setup**	Production-ready MongoDB database with correct schema design, indexes, and data relationships configured for the live environment.
6	**Production Deployment**	Live platform deployed to client-provided server: Nginx, SSL, PM2, and CI/CD pipeline configured and operational.
7	**Technical Documentation**	API endpoint reference, database schema documentation, and server environment/deployment setup guide.
8	**Client Handover Session**	60–90 minute video walkthrough demonstrating all platform features, answering client questions, and confirming delivery acceptance.

22 RISKS, ASSUMPTIONS & DEPENDENCIES

22.1 Client-Side Assumptions

- The client has an active Meta Business Account with approved Lead Ad campaigns generating leads.
- Meta API credentials (App ID, App Secret, Page Access Token) will be provided by Day 10 of the project.
- Client will provide timely feedback within 48 hours on design mockups, milestone reviews, and UAT findings.
- Client will provision the production server (minimum specification: 2 vCPU, 4GB RAM, 50GB SSD) prior to the deployment phase.
- Domain registration, DNS configuration, and associated costs are the client's responsibility.
- All platform copy, brand guidelines, and plan names/descriptions will be provided by the client in Phase 1.

22.2 Risk Register

Risk	Probability	Impact	Mitigation Strategy
Meta API credentials not provided on time	Medium	High — delays Phase 3 backend development	Flag as blocker in Day 1 kick-off. Use synthetic mock data for initial development as fallback.
Scope change requests after Phase 1 sign-off	Medium	Medium — timeline extension required	Formal change order process documented and agreed at kick-off. Phase 1 sign-off is binding.

Client feedback delays beyond 48 hours	Medium	High — cascading timeline impact	Feedback SLAs defined in kick-off. Escalation path documented. Timeline re-baselined if delay exceeds 5 days.
Distribution algorithm imbalance at scale	Low	Medium — agent fairness concern	Algorithm unit tested with simulated large lead datasets before integration. Configurable by admin.
Server performance degradation under high load	Low	High — platform reliability	Load testing conducted in Phase 5. DB indexing and query optimization completed before go-live.
Meta API breaking changes or policy updates	Low	High — integration disruption	Monitor Meta developer changelog. Webhook fallback to polling mode as resilience mechanism.
Security vulnerability in authentication	Low	Critical — data breach risk	JWT best practices strictly followed. Penetration testing checklist completed in Phase 5 before deployment.

23 COMMERCIAL TERMS & AGREEMENT

23.1 Scope Agreement

This document constitutes the complete and agreed scope of work for the Insurance Lead Management and Distribution Platform. By proceeding with this engagement, both parties confirm that the modules, features, deliverables, exclusions, and timeline defined herein represent the full project scope.

23.2 Change Management

- Any request to add, modify, or remove features after Phase 1 sign-off constitutes a formal change request.
- All change requests must be submitted in writing. Verbal agreements are not binding on either party.
- Each change request will be assessed for timeline and cost impact within 2 business days of submission.
- Development work on the change will not begin until a signed change order is received from the client.
- Minor clarifications that do not affect scope, timeline, or cost do not require a formal change order.

23.3 Confidentiality

Both parties agree to treat all information shared during this engagement — including this document, technical specifications, business data, and credentials — as strictly confidential. Neither party will disclose this information to third parties without prior written consent.